AWS Cloud Learning

DevOps Engineer · AWS Admin / SysOps

Step-by-step guide · Services to learn first · Certifications roadmap

Prepared for: DevOps Professional transitioning to AWS Cloud

Contents

1. Foundation — Common to Both Roles

These services are absolutely essential regardless of which path you choose. Master these before anything else.

Core Foundation Services — Learn These First

IAM
MUST

Identity & Access Management. Users, roles, policies, MFA, and least-privilege. The most critical AWS service — everything depends on it.

VPC
MUST

Virtual Private Cloud. Subnets, route tables, NACLs, security groups, NAT gateway, internet gateway. All AWS resources live inside a VPC.

EC2
MUST

Core compute service. Instance types, AMIs, key pairs, security groups, EBS volumes. Fundamental to understanding AWS compute.

S3
MUST

Object storage. Buckets, policies, versioning, lifecycle rules, encryption. Used by every AWS service and every role.

CloudWatch
MUST

Metrics, logs, alarms, and dashboards. The primary visibility and monitoring tool for all AWS operations.

CloudTrail
MUST

Audit log of every API call in your account. Critical for security, compliance, and investigating incidents.

Route 53
MUST

AWS DNS service. Domain management, health checks, and routing policies. Used for deployments, failover, and multi-region setups.

KMS
MUST

Key Management Service. Encryption keys for EBS, S3, RDS, and Secrets Manager. Both roles must understand encryption in AWS.

SNS
MUST

Simple Notification Service. Send email/SMS alerts, trigger Lambda, or fan-out notifications to multiple subscribers.

SQS
MUST

Simple Queue Service. Managed message queuing for decoupled architectures, retry logic, and event-driven pipelines.

Quick Tip Spend your first 2-3 weeks ONLY on IAM and VPC. Create users, assign roles, design a VPC with public/private subnets, and launch EC2 instances manually. This hands-on practice builds the intuition that makes everything else easier.

2. Learning Order — What to Start First

Follow this exact sequence. Each step builds on the previous one. Do not skip ahead — the order is carefully designed.

1

STEP 1 · Weeks 1–2

AWS Fundamentals + Account Setup

Create a free AWS account. Learn the AWS console, regions, availability zones, the shared responsibility model, and billing basics. Understand what cloud computing actually means before touching any service.

Services: AWS Console | Regions & AZs | Free Tier | Billing Dashboard

2

STEP 2 · Weeks 2–4

IAM — Identity & Access Management

This is the most important step. Learn users, groups, roles, and policies. Practice creating custom policies. Understand assume-role, MFA, and access keys. Everything in AWS is controlled by IAM — master it deeply.

Services: IAM Users | IAM Roles | IAM Policies | MFA | Access Keys

3

STEP 3 · Weeks 4–6

Networking — VPC Deep Dive

Design and build a VPC from scratch. Create public and private subnets across 2 availability zones. Add an internet gateway, NAT gateway, route tables, NACLs, and security groups. This is the backbone of all AWS architecture.

Services: VPC | Subnets | Route Tables | NAT Gateway | Security Groups | NACLs

4

STEP 4 · Weeks 6–8

Compute — EC2 + Auto Scaling

Launch EC2 instances inside your VPC. Learn instance types, AMIs, EBS volumes, and key pairs. Then set up Auto Scaling Groups with Launch Templates. Place instances behind a Load Balancer (ALB).

Services: EC2 | Auto Scaling | ALB / ELB | EBS | AMIs | Launch Templates

5

STEP 5 · Weeks 8–9

Storage — S3 + EBS + EFS

Learn S3 deeply: buckets, bucket policies, versioning, lifecycle rules, encryption, and static website hosting. Understand EBS volume types for EC2. Learn EFS for shared file storage across instances.

Services: S3 | EBS | EFS | S3 Lifecycle | Glacier

6

STEP 6 · Weeks 9–11

Databases — RDS + DynamoDB

Deploy an RDS MySQL or Postgres instance in a private subnet. Configure Multi-AZ for high availability and read replicas for performance. Then learn DynamoDB for NoSQL — understand partition keys and on-demand capacity.

Services: RDS | Aurora | DynamoDB | Multi-AZ | Read Replicas | ElastiCache

7

STEP 7 · Weeks 11–12

Monitoring — CloudWatch + CloudTrail

Set up CloudWatch metrics, log groups, and alarms. Write a CloudWatch alarm that sends an SNS notification. Enable CloudTrail and review API call logs. These are essential skills for both DevOps and Admin roles.

Services: CloudWatch | CloudTrail | SNS | EventBridge | CloudWatch Logs

8

STEP 8 · Weeks 12–15

Infrastructure as Code — CloudFormation / CDK

Write a CloudFormation template that creates a full VPC + EC2 + RDS stack. Then learn CDK to do the same in Python. IaC is non-negotiable for DevOps. Admins should also know CloudFormation for provisioning at scale.

Services: CloudFormation | CDK | SAM | Elastic Beanstalk

After completing steps 1–8 (approx. 3–4 months), split into your chosen role path on the next pages.

3. AWS DevOps Engineer — All Services

After mastering the foundation (Section 1 & 2), learn these DevOps-specific services. Focus on CI/CD pipelines, containers, and automation.

9

STEP 9 · Weeks 15–18

CI/CD Pipelines — CodePipeline + CodeBuild

Build a complete CI/CD pipeline: CodeCommit (or GitHub) → CodeBuild (compile + test) → CodeDeploy (deploy to EC2 or ECS). This is the core skill of every AWS DevOps job.

Services: CodePipeline | CodeBuild | CodeDeploy | CodeCommit | CodeArtifact

10

STEP 10 · Weeks 18–22

Containers — ECS + EKS + ECR

Learn Docker basics first, then push images to ECR. Deploy containerized apps on ECS with Fargate (serverless containers). Then learn EKS (Kubernetes on AWS) — the most in-demand DevOps skill in 2024-2025.

Services: ECS | EKS | ECR | Fargate | Docker | Kubernetes

11

STEP 11 · Weeks 22–24

Serverless — Lambda + API Gateway

Write Lambda functions in Python or Node.js. Trigger them from API Gateway, S3 events, SQS messages, or EventBridge schedules. Understand cold starts, function timeouts, and environment variables.

Services: Lambda | API Gateway | EventBridge | Step Functions | SAM

12

STEP 12 · Weeks 24–26

Security in Pipelines + Secrets

Integrate Secrets Manager into pipelines so credentials are never hardcoded. Use KMS for encrypting artifacts and environment variables. Understand WAF, Shield, and how to scan containers for vulnerabilities.

Services: Secrets Manager | KMS | WAF | ECR Scanning | IAM Roles for Tasks

All DevOps Services — Quick Reference

CI/CD & Automation

CodePipeline
MUST

Orchestrates full CI/CD pipeline from source to deploy.

CodeBuild
MUST

Managed build/test service. Runs in containers.

CodeDeploy
MUST

Automates deployments to EC2, ECS, Lambda.

CodeCommit
GOOD

AWS-managed Git repo. Often replaced by GitHub.

CodeArtifact
GOOD

Artifact repo for npm, Maven, pip packages.

IaC & Deployment

CloudFormation
MUST

AWS-native IaC with JSON/YAML templates.

CDK
MUST

Write IaC in Python/TypeScript. Modern approach.

Elastic Beanstalk
GOOD

PaaS for simpler apps. Handles deployments.

SAM
GOOD

Simplified IaC for serverless apps.

Containers & Serverless

ECS
MUST

Run Docker containers with Fargate or EC2 launch type.

EKS
MUST

Managed Kubernetes — most in-demand DevOps skill.

ECR
MUST

Container image registry. Scan images for CVEs.

Lambda
MUST

Event-driven serverless functions.

Fargate
GOOD

Serverless compute engine for ECS/EKS.

Monitoring & Tracing

CloudWatch
MUST

Metrics, logs, alarms, dashboards. Core observability.

X-Ray
MUST

Distributed tracing for microservices and serverless.

CloudTrail
MUST

API audit trail for security and compliance.

EventBridge
MUST

Event bus for automation and cross-service triggers.

Config
GOOD

Track config drift. Enforce compliance rules.

Networking & CDN

ALB / ELB

MUST

Load balance traffic across EC2, containers, Lambda.

CloudFront

GOOD

CDN for static files and APIs.

API Gateway

GOOD

Managed REST/HTTP/WebSocket API endpoints.

4. AWS Admin / SysOps — All Services

After the foundation, AWS Admins focus on operations, governance, security, cost management, and keeping production environments healthy.

9

STEP 9 · Weeks 15–17

Systems Manager — Fleet Management

AWS Systems Manager (SSM) is the admin's most powerful tool. Use Session Manager to access EC2 without SSH. Use Patch Manager to patch hundreds of servers. Store configuration in Parameter Store.

Services: SSM Session Manager | Patch Manager | Parameter Store | Run Command

10

STEP 10 · Weeks 17–20

Multi-Account Governance

Learn AWS Organizations to manage multiple accounts centrally. Apply Service Control Policies (SCPs) to restrict what accounts can do. Use Control Tower to set up new accounts with pre-built guardrails.

Services: AWS Organizations | Control Tower | SCPs | Account Factory

11

STEP 11 · Weeks 20–23

Security & Threat Detection

Enable GuardDuty for intelligent threat detection across all accounts. Set up Security Hub to aggregate findings. Use Macie to find sensitive data in S3. These are now mandatory in most enterprise AWS environments.

Services: GuardDuty | Security Hub | Macie | Inspector | Detective

12

STEP 12 · Weeks 23–25

Cost Management & Optimization

Use Cost Explorer to identify top cost drivers. Set Budgets with alert thresholds. Run Trusted Advisor to find waste. Learn Reserved Instances and Savings Plans for 40-70% cost reduction on committed workloads.

Services: Cost Explorer | Budgets | Trusted Advisor | Reserved Instances | Savings Plans

All Admin Services — Quick Reference

Compute & Scaling

EC2 + Auto Scaling
MUST

Instance lifecycle, ASGs, launch templates, scaling policies.

Systems Manager
MUST

Patch, run commands, session access, parameter store.

Lightsail
GOOD

Simple compute for small workloads.

Networking & Connectivity

Direct Connect
MUST

Private dedicated line from on-premises to AWS.

VPN (Site-to-Site)
MUST

Encrypted tunnel between on-prem and AWS VPC.

Transit Gateway
MUST

Hub connecting multiple VPCs and on-prem networks.

Global Accelerator
GOOD

Improve availability using AWS backbone network.

Storage & Backup

EBS
MUST

Block storage. Snapshots, encryption, IOPS tuning.

EFS
MUST

Shared NFS file system across EC2 instances.

AWS Backup
MUST

Centralized backup for EC2, RDS, EFS, DynamoDB.

Storage Gateway
GOOD

Hybrid bridge between on-premises and AWS storage.

Glacier
GOOD

Long-term archival storage via S3 lifecycle.

Identity, Security & Compliance

Organizations
MUST

Manage multiple accounts, billing, and SCPs.

Control Tower
MUST

Govern multi-account setup with guardrails.

Security Hub
MUST

Central security findings and compliance checks.

GuardDuty
MUST

ML-based threat detection on logs and traffic.

Macie
GOOD

Discover and protect sensitive data in S3.

Inspector
GOOD

Vulnerability assessment for EC2 and containers.

Cost & Governance

Cost Explorer
MUST

Visualize and analyze AWS costs over time.

Budgets
MUST

Set spending alerts before costs exceed targets.

Trusted Advisor
MUST

Real-time recommendations across 5 categories.

Service Catalog
GOOD

Manage approved IT service portfolios.

Resource Groups
GOOD

Organize resources by tags for bulk operations.

Monitoring & Operations

Config
MUST

Track config changes. Enforce compliance rules.

Health Dashboard
MUST

See AWS service events affecting your account.

OpsWorks
GOOD

Config management with Chef/Puppet.

Databases

RDS + Aurora
MUST

Managed relational DBs. Multi-AZ, backups, maintenance.

DynamoDB
GOOD

Manage capacity, global tables, backups, and DAX.

ElastiCache
GOOD

Manage Redis/Memcached clusters for caching.

5. Certification Roadmap

AWS certifications are globally recognized and significantly boost your salary and job opportunities. Follow this order.

1

AWS Cloud Practitioner (CCP)

Exam code: CLF-C02 · Study time: 1–2 months

Start here. Covers cloud concepts, AWS services overview, pricing, support plans, and the shared responsibility model. No prior cloud experience needed. Gives you the vocabulary and big-picture understanding for all future certs.

2

AWS Solutions Architect Associate

Exam code: SAA-C03 · Study time: 2–3 months

The most popular AWS cert. Covers VPC, EC2, S3, RDS, IAM, high availability, disaster recovery, and architecture best practices. This is the BEST foundation for both DevOps and Admin paths — do this before the specialty certs.

3A

DevOps Engineer Professional

Exam code: DOP-C02 · Study time: 3–4 months

TARGET CERT for DevOps path. Covers CI/CD, IaC, monitoring, deployment strategies (blue/green, canary), and incident response. Requires Solutions Architect Associate or SysOps Associate as prerequisite. Highly valued in the industry.

3B

SysOps Administrator Associate

Exam code: SOA-C02 · Study time: 2–3 months

TARGET CERT for Admin path. Covers monitoring, metrics, automation, backups, networking, cost management, and security. Hands-on exam with a practical lab section. Can be taken alongside or before the DevOps Professional cert.

4

AWS Security Specialty

Exam code: SCS-C02 · Study time: 3–4 months

Highly recommended for both paths after getting your primary cert. Covers IAM deep dive, encryption, threat detection, incident response, and compliance. Security skills command premium salaries.

6. Study Resources & Tips

The best platforms and study strategies for learning AWS efficiently.

1

Stephane Maarek — Udemy Courses

The most recommended AWS instructor worldwide. His Solutions Architect, DevOps, and SysOps courses are extremely clear and regularly updated. Watch at 1.25x speed. Buy only during Udemy sales (under $15).

Link: udemy.com/user/stephane-maarek

2

AWS Skill Builder (Free)

AWS's official learning platform. Has free digital courses for every service, official exam prep, and hands-on labs. Create a free account at skillbuilder.aws. The 'AWS Cloud Quest' gamified learning path is excellent for beginners.

Link: skillbuilder.aws

3

A Cloud Guru / Pluralsight

Video courses with built-in cloud sandbox environments. Great for hands-on practice without needing your own AWS account. Has learning paths for DevOps and SysOps.

Link: acloudguru.com

4

AWS Free Tier Account

Create your own AWS account and use the Free Tier (12 months). Practice every service hands-on. Always set a Budget alert at $10 to avoid surprise bills. Hands-on practice is 10x more valuable than watching videos alone.

Link: aws.amazon.com/free

5

TutorialsDojo Practice Exams

The best practice exam questions for all AWS certifications. After finishing a course, do 3-5 full practice exams. Aim for 85%+ before booking the real exam. Each wrong answer has a detailed explanation — read every one.

Link: tutorialsdojo.com

Study Plan Summary — Month by Month

Phase	Time	Key Services	Certification Goal
AWS Fundamentals	Month 1	Console, Regions, AZs, Billing, Free Tier	**Foundation**
IAM + VPC + EC2	Month 2	IAM, VPC, Subnets, NACLs, Security Groups, EC2, EBS	**Cloud Practitioner CLF-C**
Core Services	Month 3	S3, RDS, CloudWatch, CloudTrail, SNS, SQS, Route 53	**SAA-C03 Prep**
Architecture	Month 4	CloudFormation, CDK, ECS, Lambda, API Gateway, CloudFront	**Solutions Architect SAA-**
DevOps Path	Month 5-7	CodePipeline, CodeBuild, EKS, ECR, X-Ray, Secrets Manager	**DevOps Pro DOP-C02**

Admin Path	Month 5-6	SSM, Organizations, Control Tower, GuardDuty, Cost Explorer	**SysOps Admin SOA-C02**
Security Advanced	Month 7+	KMS, Macie, Inspector, WAF, Shield, Detective, ACM	**Security Specialty SCS-C02**

Top Study Tips for AWS Success

1 **Hands-on beats videos every time**

For every concept you watch or read, immediately try it in the AWS console. Set a $10 budget alert on your Free Tier account so you never get surprise bills.

2 **Learn IAM until it feels natural**

Most AWS interview questions and real-world problems come down to IAM. Spend double the time here compared to any other service.

3 **Use AWS documentation as your second textbook**

Every AWS service has a 'Getting Started' guide and FAQs. Reading these builds understanding that video courses miss.

4 **Practice exam questions are mandatory**

Do TutorialsDojo practice exams AFTER finishing your course. Aim for 85%+ consistently before booking the real exam. Read every explanation.

5 **Build real projects, not just labs**

Deploy a simple web application with EC2 + RDS + S3 + ALB. Then automate its deployment with CodePipeline. This one project teaches more than 20 labs.

6 **Join the community**

Reddit r/AWSCertifications and r/aws are excellent for questions and motivation. The AWS community is very welcoming to beginners.